

SCOPE OF WORK (SOW)

Enterprise Property & Hotel Management System (PMS)

Software Development Project

Prepared for the Development Team

Document Version 1.0

Date: _____

Prepared by: _____

Document Control

Revision History

Version	Description
v1.0	Initial draft of complete Scope of Work

Purpose of this Document

This Scope of Work (SOW) explains, in plain and simple language, exactly what needs to be built for the Property/Hotel Management System (PMS). It lists every module, every feature inside that module, and the detailed parameters/rules the developers must follow. It also explains the technology expectations, timeline, testing process, and what "done" looks like for each part of the system. Developers should treat this document as the single source of truth for what to build.

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1. Introduction & Project Overview

1.1 What We Are Building

We are building a complete, enterprise-grade Property/Hotel Management System (PMS) — a web-based (and mobile-supported) software platform that lets a hotel, resort, or multi-property group run its entire daily operation from one system. This includes taking bookings, checking guests in and out, managing rooms and rates, billing guests, running restaurants/bars (POS), managing housekeeping and maintenance staff, connecting to booking websites like Booking.com and Expedia, and giving management real-time reports on how the business is performing.

1.2 Who Will Use This System

- Front Desk / Reception staff — for check-in, check-out, and guest requests
- Reservation Team — for creating, modifying, and confirming bookings
- Housekeeping Staff & Supervisors — for room cleaning and status updates
- Maintenance Team — for repair and upkeep requests
- Restaurant / Bar / Spa Staff — for point-of-sale billing
- Finance / Accounts Team — for invoicing, payments, and reconciliation
- Sales & Marketing Team — for corporate rates, packages, and guest loyalty
- General Manager / Owners — for dashboards, reports, and business decisions
- System Administrator — for user access, configuration, and security
- Guests — through a guest-facing booking portal and/or mobile app

1.3 Business Goals of the Project

- Replace manual/paper or outdated systems with one unified digital platform
- Reduce booking errors (double-bookings, wrong rates) to near zero
- Give management real-time visibility into occupancy, revenue, and staff performance
- Automate repetitive tasks (night audit, invoicing, reminders, reports)
- Support growth from a single property to a multi-property/multi-brand group
- Meet enterprise standards for security, uptime, and data protection

2. Project Scope

2.1 In-Scope (What Developers MUST Build)

The list below is the full set of modules included in this project. Each one is explained in complete detail in Section 4 (Functional Modules).

- Reservation & Booking Engine
- Front Desk Operations (Check-in / Check-out)
- Room, Rate & Inventory Management
- Revenue & Dynamic Pricing Management
- Housekeeping Management
- Point of Sale (POS) for Restaurant, Bar, Spa & Mini-Bar
- Billing, Invoicing, Folio & Payments
- Guest CRM & Loyalty Program
- Channel Manager & OTA Integration
- Banquet, Conference & Event Management
- Maintenance / Engineering Management
- HR & Staff Management (Attendance, Shifts, Payroll hooks)
- Reporting, Analytics & Management Dashboard
- Multi-Property / Multi-Brand Management
- User Roles, Permissions & System Administration
- Notifications Engine (Email, SMS, Push, WhatsApp)
- Guest-Facing Web Booking Portal
- Mobile Applications (Guest App and Staff App)

2.2 Out-of-Scope (Not Included Unless Separately Agreed)

- Physical hardware procurement (POS terminals, door-lock hardware, card printers) — software integration only
- Custom accounting/ERP software — the system will integrate with existing accounting tools, not replace them
- Physical network/Wi-Fi infrastructure setup at the property
- Content writing/photography for the guest booking website beyond placeholder content
- Any government licensing, tax registration, or legal compliance filing (system will support the data needed, but filing is the client's responsibility)

3. System Architecture & Technology Approach

3.1 Architecture Overview

The system must be built as a cloud-based, multi-tenant capable, modular application. "Modular" means each part described in Section 4 (Reservations, POS, Housekeeping, etc.) should work as its own service or clearly separated module, so it can be updated, scaled, or fixed without breaking the rest of the system. "Multi-tenant capable" means the same software can serve many hotel properties, each with its own data kept completely separate and secure.

3.2 Recommended Technical Approach

Parameter	Description / Requirement
Architecture Style	Modular monolith or microservices, exposed through a documented REST/GraphQL API layer so the web app, mobile apps, and any future third party can all use the same backend.
Frontend (Web)	Modern JavaScript framework (e.g., React, Vue, or Angular) with a responsive design that works on desktop, tablet, and mobile browsers.
Mobile Apps	Cross-platform framework (e.g., React Native or Flutter) for both the Guest App and the Staff App, or native iOS/Android if the client requires it.
Backend	A scalable server-side framework (e.g., Node.js, Java Spring Boot, .NET, or Python/Django) chosen based on the development team's expertise.
Database	A relational database (e.g., PostgreSQL or MySQL) for transactional data (bookings, billing), with an optional NoSQL store (e.g., MongoDB/Redis) for caching, logs, and session data.
Hosting	Cloud infrastructure (e.g., AWS, Azure, or Google Cloud) with auto-scaling, load balancing, and daily automated backups.
API Design	RESTful APIs with versioning (e.g., /api/v1/), documented using OpenAPI/Swagger so third-party systems (OTAs, payment gateways) can integrate easily.
Real-Time Updates	WebSocket or similar technology so room status, bookings, and dashboards update live across all connected users without needing a page refresh.
Offline Support	Front desk and POS modules should keep working (in a limited mode) during short internet outages and automatically sync once the connection returns.

4. Functional Modules — Full Detail

This is the main part of the document. Every module below must be built with all the listed features and parameters. Developers should read each module fully before starting development, since many modules depend on data or rules set up in another module (for example, Billing depends on Room & Rate setup).

Module 1: Reservation & Booking Engine

This module is the heart of the system. It allows staff (and guests, through the booking portal) to search room availability and create, modify, or cancel bookings.

Key Features

- Real-time room availability search by date, room type, and number of guests
- Individual, group, and corporate bookings
- Multiple rate plans per room type (e.g., Refundable, Non-Refundable, Breakfast Included)
- Booking hold/waitlist when rooms are full
- Split bookings and room-move handling
- Deposit and advance payment collection at time of booking
- Automatic booking confirmation email/SMS with a unique reservation ID
- Cancellation and no-show handling with configurable penalty rules
- Overbooking control with manager override and audit trail
- Calendar/tape chart view showing all rooms and bookings visually

Detailed Parameters & Requirements

Parameter	Description / Requirement
Booking Statuses	Tentative, Confirmed, Checked-In, Checked-Out, Cancelled, No-Show, Waitlisted
Booking Sources	Walk-in, Phone, Website, OTA, Travel Agent, Corporate, Mobile App
Rate Types	Best Available Rate (BAR), Corporate Rate, Package Rate, Promotional Rate, Group Rate
Minimum Data per Booking	Guest name, contact number, email, ID/passport number, arrival date, departure date, room type, number of adults/children, rate plan, payment method
Cancellation Rules	Configurable per rate plan — e.g., free cancellation up to 24/48/72 hours before arrival, else charge one night or full amount
Group Booking Rules	Ability to block 10+ rooms under one master booking with individual guest names added later (rooming list)
Booking Modification Rules	Date change, room change, and rate change must be logged with timestamp and staff name for audit purposes

Module 2: Front Desk Operations (Check-In / Check-Out)

This module supports everything the reception desk does on a daily basis, from the moment a guest walks in to the moment they leave.

Key Features

- Quick check-in (search by name, confirmation number, or phone number)
- ID/passport scanning and storage (photo upload supported)
- Digital registration card with e-signature capture
- Room key/card activation trigger (integration-ready for electronic door locks)
- Early check-in and late check-out handling with optional charges
- Walk-in guest booking directly from the front desk screen
- Express check-out (auto-generate final bill and email it to the guest)
- Night Audit process to close each day's transactions automatically
- Guest message and request logging (e.g., wake-up call, extra towels)

Detailed Parameters & Requirements

Parameter	Description / Requirement
Check-in Requirements	Government ID capture, guest signature, deposit/payment confirmation, room assignment
Night Audit Tasks	Roll over system date, post room charges & taxes for the night, close cashier shifts, generate daily revenue report
Room Assignment Logic	Auto-assign based on room type, floor/view preference, housekeeping status (must be "Clean & Inspected"), and VIP flags
Early/Late Fees	Configurable percentage or flat fee, auto-applied to folio when triggered
Data Retention	Guest ID documents must be stored securely and only accessible to authorized front-desk and compliance roles

Module 3: Room, Rate & Inventory Management

This module is where the hotel sets up its physical rooms, room types, and how those rooms are priced and made available for sale.

Key Features

- Setup of room types (e.g., Standard, Deluxe, Suite) with photos, amenities, and max occupancy
- Individual room setup (room number, floor, view, connecting rooms, accessibility features)
- Rate plan builder with seasonal, weekday/weekend, and length-of-stay pricing
- Inventory allocation per channel (e.g., 5 rooms reserved for the website, 3 for OTAs)
- Room status board (Vacant Clean, Vacant Dirty, Occupied, Out of Order, Out of Service)
- Block/unblock rooms for maintenance with reason tracking

Detailed Parameters & Requirements

Parameter	Description / Requirement
Room Attributes	Room number, type, floor, max occupancy (adults/children), bed configuration, smoking/non-smoking, accessible room flag
Rate Rule Options	Minimum Length of Stay (MinLOS), Maximum Length of Stay (MaxLOS), Closed to Arrival (CTA), Closed to Departure (CTD), blackout dates
Inventory Sync	Availability changes must reflect across all sales channels within 1 minute to avoid overbooking
Out-of-Order Rules	Rooms marked Out of Order/Out of Service must automatically be removed from bookable inventory and flagged to Maintenance

Module 4: Revenue & Dynamic Pricing Management

This module helps management maximize income by adjusting prices based on demand, competitor pricing, and occupancy levels.

Key Features

- Dashboard showing occupancy forecast vs. current pricing
- Rule-based dynamic pricing (e.g., raise rate by 10% when occupancy exceeds 80%)
- Competitor rate tracking (manual entry or integration with a rate-shopping tool)
- Package creation (e.g., Room + Breakfast + Spa bundled at one price)
- Promo code / discount voucher engine

Detailed Parameters & Requirements

Parameter	Description / Requirement
Pricing Triggers	Occupancy percentage thresholds, day of week, special events/holidays, booking lead time
Approval Levels	Automatic pricing changes below a set % require no approval; changes above threshold require manager approval
Promo Code Rules	Fixed amount or percentage discount, usage limits, expiry date, applicable room types/rate plans

Module 5: Housekeeping Management

This module coordinates the cleaning and readiness of every room, connecting the housekeeping team with the front desk in real time.

Key Features

- Room status updates by housekeeping staff via mobile/tablet (Dirty, Cleaning, Clean, Inspected)
- Automatic task assignment to housekeeping staff based on checkouts and floor zones
- Supervisor inspection and approval workflow before a room becomes bookable
- Lost & found item logging linked to room and date

- Linen, amenities, and minibar restocking checklist per room
- Do Not Disturb (DND) flagging synced from guest room controls or manual entry

Detailed Parameters & Requirements

Parameter	Description / Requirement
Room Status Flow	Dirty → Cleaning in Progress → Clean → Inspected → Ready for Guest
Task Assignment Logic	Auto-assign by floor/zone and staff availability; supervisors can manually reassign
Turnaround Time Target	Configurable target (e.g., 30 minutes per standard room) with alerts if exceeded
Deep-Clean Scheduling	Ability to schedule recurring deep cleaning (e.g., every 30 days) independent of checkout

Module 6: Point of Sale (POS) — Restaurant, Bar, Spa & Mini-Bar

This module allows outlets inside the property (restaurant, bar, spa, gift shop, mini-bar) to sell items and services, with the option to charge them directly to the guest's room.

Key Features

- Menu/item and price list management per outlet
- Order taking by table/room with item modifiers (e.g., extra cheese, no ice)
- Split bill and merge bill functionality
- "Charge to Room" option that posts directly to the guest folio
- Kitchen Order Ticket (KOT) printing/routing to kitchen display
- Inventory deduction linked to recipe/ingredient usage (optional stock module)
- Tips and service charge handling

Detailed Parameters & Requirements

Parameter	Description / Requirement
Outlet Types Supported	Restaurant, Bar, Spa/Wellness, Gift Shop, Mini-Bar, Room Service, Laundry
Payment Options at POS	Cash, Card, Room Charge, Wallet/Points redemption
Tax Handling	Support multiple tax types (e.g., service tax, VAT/GST) applied per outlet or item category
Charge-to-Room Validation	System must check the guest's credit limit/authorization before allowing a room charge

Module 7: Billing, Invoicing, Folio & Payments

This module manages all money-related transactions for a guest's stay, from the first charge to the final receipt.

Key Features

- Guest folio (running bill) that automatically collects room charges, taxes, and outlet charges
- Split folio (e.g., room charges to company, incidentals to personal card)
- Multiple payment methods per booking (part cash, part card, part company bill)
- Invoice and receipt generation (PDF, printable, and emailable)
- Refund processing with approval workflow
- Integration-ready payment gateway connection for online and in-person card payments
- City ledger / accounts receivable for corporate clients who pay later
- Currency conversion support for international guests

Detailed Parameters & Requirements

Parameter	Description / Requirement
Folio Types	Master Folio (whole stay), Split Folio, Group Master Folio with individual sub-folios
Payment Methods	Cash, Credit/Debit Card, Bank Transfer, Online Payment Gateway, Wallet, Loyalty Points, Corporate Direct Billing
Tax Compliance	Configurable tax rates and rules by region/country, tax-inclusive and tax-exclusive pricing options
Refund Rules	Refunds above a set amount require manager or finance-head approval before processing
Invoice Numbering	Sequential, non-repeating, and compliant with local invoicing/tax regulations

Module 8: Guest CRM & Loyalty Program

This module keeps a complete history and profile of every guest, so the hotel can build long-term relationships and offer personalized service.

Key Features

- Guest profile with stay history, preferences, and special requests
- Guest tagging (VIP, Blacklisted, Repeat Guest, Corporate)
- Loyalty points earning and redemption rules
- Membership tiers (e.g., Silver, Gold, Platinum) with tier-based perks
- Birthday/anniversary and special-occasion reminders for personalized offers
- Feedback and review collection after check-out
- Merge duplicate guest profiles

Detailed Parameters & Requirements

Parameter	Description / Requirement
Guest Profile Fields	Name, contact details, ID number, nationality, preferences (room type, floor, dietary), stay history, total spend
Loyalty Points Logic	Points earned per currency unit spent; configurable earn-rate and redemption-rate per tier
Tier Upgrade Rules	Automatic tier upgrade/downgrade based on nights stayed or amount spent within a rolling 12-month period
Data Privacy	Guest data must be handled per applicable privacy laws (e.g., GDPR) — see Non-Functional Requirements

Module 9: Channel Manager & OTA Integration

This module connects the hotel's room inventory and rates to third-party booking websites (OTAs) like Booking.com, Expedia, Airbnb, and Agoda, keeping everything updated automatically so the hotel never gets double-booked.

Key Features

- Two-way sync of room availability and rates with connected OTAs
- Automatic booking import from OTAs directly into the Reservation module
- Rate parity monitoring (alerts if a rate on an OTA doesn't match the agreed rate)
- Commission tracking per OTA booking
- Global Distribution System (GDS) connectivity for travel-agent bookings (optional/enterprise)
- Single dashboard to manage all channels instead of logging into each OTA extranet separately

Detailed Parameters & Requirements

Parameter	Description / Requirement
Sync Frequency	Real-time or near real-time (within 1–5 minutes) push/pull of rates and availability
Supported Channels (minimum)	Booking.com, Expedia, Agoda, Airbnb, Hotels.com — expandable via channel manager partner APIs
Conflict Handling	If two channels try to book the last room simultaneously, the system must lock inventory and reject the second booking automatically, notifying the OTA
Commission Field	Each OTA booking must store the commission percentage/amount for accurate net revenue reporting

Module 10: Banquet, Conference & Event Management

For properties with event spaces, this module manages the booking and billing of banquet halls, meeting rooms, and event catering.

Key Features

- Event space inventory and availability calendar

- Event booking with room layout, catering, and equipment requirements
- Quotation and contract generation for corporate/event clients
- Event-specific billing linked to the main billing module
- Banquet Event Order (BEO) generation for kitchen/service staff

Detailed Parameters & Requirements

Parameter	Description / Requirement
Event Space Attributes	Name, capacity (theatre/classroom/banquet style), floor, hourly/daily rate, available equipment
Booking Data	Client name, event type, date/time, guest count, catering menu, setup requirements, payment terms
Approval Workflow	Quotations above a configurable value require sales-manager approval before confirmation

Module 11: Maintenance / Engineering Management

This module tracks repair and upkeep tasks for rooms, equipment, and common areas, ensuring nothing gets forgotten.

Key Features

- Maintenance ticket creation (by any staff member) linked to a room or asset
- Priority levels and assignment to engineering staff
- Preventive maintenance scheduling (e.g., AC servicing every 3 months)
- Ticket status tracking from Open to Resolved with photo attachment support
- Automatic room block/unblock when a ticket affects room availability

Detailed Parameters & Requirements

Parameter	Description / Requirement
Ticket Priority Levels	Low, Medium, High, Urgent (Urgent tickets trigger immediate notification to engineering supervisor)
Asset Register	Each major asset (AC unit, elevator, boiler, etc.) should have its own maintenance history log
SLA Targets	Configurable resolution time targets per priority level, with escalation if breached

Module 12: HR & Staff Management

This module manages basic staff administration so operational modules (housekeeping, front desk, POS) know who is working and when.

Key Features

- Staff profile and role assignment
- Shift scheduling and roster management
- Attendance/clock-in-clock-out tracking
- Leave request and approval workflow
- Basic performance tracking (tasks completed, guest feedback linked to staff)
- Export-ready data for payroll systems (not a full payroll engine)

Detailed Parameters & Requirements

Parameter	Description / Requirement
Staff Data Fields	Name, employee ID, department, designation, shift pattern, contact details, joining date
Shift Types	Morning, Afternoon, Night, Split Shift — fully configurable per property
Attendance Method	Manual entry, biometric integration-ready, or mobile app geofenced check-in

Module 13: Reporting, Analytics & Management Dashboard

This module turns all the data collected across the system into clear reports and visual dashboards so managers and owners can make informed decisions quickly.

Key Features

- Real-time dashboard: occupancy %, ADR (Average Daily Rate), RevPAR (Revenue Per Available Room), today's arrivals/departures
- Financial reports: daily revenue, outstanding payments, tax summary, outlet-wise sales
- Operational reports: housekeeping performance, maintenance turnaround, staff attendance
- Guest reports: repeat guest ratio, loyalty program performance, feedback scores
- Custom report builder with export to Excel/PDF
- Scheduled email reports (daily/weekly/monthly) to management

Detailed Parameters & Requirements

Parameter	Description / Requirement
Core KPIs	Occupancy %, ADR, RevPAR, GOPPAR, Total Revenue, Outstanding Balance, Cancellation Rate, No-Show Rate
Report Formats	On-screen dashboard, PDF export, Excel/CSV export
Access Control	Financial reports restricted to Management/Finance roles only (see Module 15)
Data Refresh Rate	Dashboards must refresh in real time or at most every 5 minutes

Module 14: Multi-Property / Multi-Brand Management

For hotel groups with more than one property, this module allows central control while keeping each property's data separate and secure.

Key Features

- Central dashboard showing all properties' performance side by side
- Property-level configuration (own rooms, rates, staff, and branding) under one group account
- Central reservation option (book any property from one screen)
- Group-wide reporting rolled up from individual properties
- Role-based access — e.g., a General Manager only sees their property, while group owners see all

Detailed Parameters & Requirements

Parameter	Description / Requirement
Data Isolation	Each property's guest, financial, and staff data must be logically separated (multi-tenant architecture)
Central Configuration Items	Brand standards, loyalty program rules, tax templates can be pushed from head office to all properties
Cross-Property Reporting	Ability to compare occupancy/revenue across properties in one report

Module 15: User Roles, Permissions & System Administration

This module controls who can see and do what inside the system — a critical requirement for an enterprise-grade product.

Key Features

- Role-based access control (RBAC) with predefined and custom roles
- Granular permission settings (view-only, edit, approve, delete) per module
- Full audit trail/log of every important action (who did what, and when)
- Two-factor authentication (2FA) for admin and finance roles
- System configuration panel (tax rates, currencies, languages, property details, email templates)
- User activity monitoring and session management (auto-logout after inactivity)

Detailed Parameters & Requirements

Parameter	Description / Requirement
Default Roles (minimum)	System Administrator, General Manager, Front Office Manager, Front Desk Agent, Housekeeping Supervisor/Staff, POS Cashier, Finance/Accounts, Sales & Marketing, Maintenance/Engineering
Permission Granularity	Each role must be configurable at the module and action level (e.g., can view billing but cannot issue refunds)
Audit Log Fields	User, action performed, date/time, module affected, before/after values where applicable

Parameter	Description / Requirement
Password Policy	Minimum length and complexity rules, forced periodic reset, lockout after repeated failed attempts

Module 16: Notifications Engine

This module makes sure guests and staff are informed automatically at the right moments, without manual effort.

Key Features

- Automated guest emails/SMS: booking confirmation, pre-arrival reminder, check-in instructions, post-stay thank-you/feedback request
- Staff notifications: new booking alert, maintenance ticket assigned, low inventory alert, VIP arrival alert
- WhatsApp/SMS gateway integration option
- Customizable message templates per property and per language
- Notification history log per guest and per staff member

Detailed Parameters & Requirements

Parameter	Description / Requirement
Trigger Events	Booking created, booking modified, booking cancelled, check-in, check-out, payment received, ticket assigned, low stock
Channels Supported	Email, SMS, Push Notification (mobile app), WhatsApp Business API (optional)
Template Management	Admins must be able to edit templates without needing a developer, using placeholders like {GuestName}, {CheckInDate}

Module 17: Guest-Facing Web Booking Portal

A public website/booking page where guests can search availability, view rates, and book directly — reducing dependency on OTA commissions.

Key Features

- Room search with date, guest count, and room-type filters
- Secure online payment for booking deposit or full payment
- Best-rate guarantee messaging and promo code entry
- Mobile-responsive design
- Guest account creation to view/manage their own bookings

Detailed Parameters & Requirements

Parameter	Description / Requirement
Page Load Target	Under 3 seconds on a standard broadband/4G connection
Payment Security	Must comply with PCI-DSS standards (see Non-Functional Requirements)

Parameter	Description / Requirement
Language/Currency	Multi-language and multi-currency display, configurable per property

Module 18: Mobile Applications (Guest App & Staff App)

Two separate mobile apps: one for guests to enhance their stay, and one for staff to manage operations on the move.

Key Features

- Guest App: view/manage booking, digital check-in, room service ordering, in-app chat with front desk, loyalty points tracker
- Staff App: housekeeping task list, maintenance ticket updates, front-desk mobile check-in, manager dashboard on the go, push notifications for urgent tasks
- Offline-capable task lists for housekeeping/maintenance staff with auto-sync

Detailed Parameters & Requirements

Parameter	Description / Requirement
Platforms	iOS and Android (via cross-platform framework recommended in Section 3)
Guest App Login	Booking reference + last name, or full account login with email/phone
Staff App Login	Same credentials as the main system, restricted by role as defined in Module 15

5. Third-Party Integrations

An enterprise PMS rarely works alone — it must talk to other systems. The table below lists every integration the developers should plan for, along with why it's needed.

Integration	Purpose
Payment Gateways	To accept online and in-person card/digital payments securely (e.g., Stripe, Razorpay, PayU, or local providers as required).
Channel Manager / OTAs	To sync availability and rates with Booking.com, Expedia, Airbnb, Agoda, etc. (see Module 9).
Electronic Door Lock Systems	To generate and activate digital room keys at check-in (integration hooks only; hardware is out of scope).
Accounting / ERP Software	To export invoices, revenue, and tax data to the client's existing accounting system (e.g., QuickBooks, Tally, SAP).
Email & SMS Gateways	To send automated guest and staff notifications (see Module 16).
ID Verification / Government Portals	Where legally required, to submit guest ID data to local police/tourism authorities automatically.
Review & Reputation Platforms	To pull in or push out guest reviews (e.g., Google Reviews, TripAdvisor) for the CRM/feedback module.
GDS (Global Distribution System)	Optional, for large properties relying on travel-agent bookings via Amadeus, Sabre, or Travelport.
Biometric/Access Control	Optional integration hooks for staff attendance and secure-area access.

6. Non-Functional Requirements

These are the "behind the scenes" requirements that make the system truly enterprise-grade. They are just as important as the features themselves.

6.1 Security

- All data must be encrypted in transit (HTTPS/TLS) and at rest (database-level encryption)
- Payment processing must be PCI-DSS compliant — card details should never be stored directly on our servers unless using a certified vault
- Role-based access control and two-factor authentication for sensitive roles (see Module 15)
- Regular security audits and vulnerability scanning before go-live and periodically after
- Protection against common web attacks (SQL injection, cross-site scripting, CSRF)

6.2 Performance

- Standard pages/screens must load within 3 seconds under normal load
- The system must support at least the number of concurrent users agreed for the property size (to be confirmed with the client, but design for at least 200 concurrent users as a baseline for a mid-size enterprise deployment)
- Search/availability queries must return results within 2 seconds

6.3 Scalability & Reliability

- Architecture must support adding new properties or modules without a full system rebuild
- Target system uptime of 99.9% (excluding scheduled maintenance windows)
- Automated daily backups with a tested disaster-recovery/restore process

6.4 Compliance & Data Privacy

- Compliance with applicable data protection laws (e.g., GDPR for European guests, or local equivalents)
- Guests must be able to request a copy of, or deletion of, their personal data where legally required
- Clear data retention policy for guest ID documents and financial records

6.5 Usability & Accessibility

- Interface must be simple enough for staff with basic computer literacy to learn within a few hours
- Support for multiple languages (at minimum, English plus one local language, expandable)
- Design should follow accessibility best practices (readable fonts, sufficient color contrast, keyboard navigation)

6.6 Auditability

- Every financial transaction, rate change, and cancellation must be logged with user, timestamp, and reason
- Reports must be reproducible — running the same report for the same date range should always give the same result

7. Development Methodology & Project Phases

7.1 Recommended Methodology

Development should follow an Agile (Scrum) approach, working in 2-week sprints, with each module developed, tested, and demonstrated to the client incrementally rather than all at once at the end. This reduces risk and allows early feedback.

7.2 Suggested Phases

Phase	What Happens
Phase 1: Discovery & Design	Requirement confirmation, database design, UI/UX wireframes and prototypes, technical architecture finalization, and sign-off on this SOW.
Phase 2: Core Foundation	User authentication, roles & permissions, property/room setup, and the base admin panel that every other module depends on.
Phase 3: Core Operations	Reservation Engine, Front Desk (Check-in/Check-out), Room & Rate Management, Housekeeping.
Phase 4: Revenue & Billing	Billing/Folio/Payments, Revenue & Dynamic Pricing, Point of Sale (POS).
Phase 5: Guest Experience & Distribution	Guest CRM & Loyalty, Channel Manager/OTA Integration, Guest Web Booking Portal.
Phase 6: Extended Modules	Banquet/Event Management, Maintenance, HR & Staff Management.
Phase 7: Intelligence & Mobile	Reporting/Analytics Dashboard, Multi-Property Management, Guest App and Staff App.
Phase 8: Integration, Testing & Hardening	All third-party integrations (Section 5), full QA testing cycle, security testing, performance/load testing.
Phase 9: Deployment & Training	Production deployment, data migration (if applicable), staff training, and go-live support.
Phase 10: Post-Launch Support	Hypercare period, bug fixing, and transition into the ongoing support/maintenance agreement.

Note: Exact durations/dates should be filled in by the project manager once team size and priorities are confirmed. Phases 2–7 can run partially in parallel across different developer teams since modules are built to be independent.

8. Deliverables

At the end of the project, the development team must hand over the following:

1. Fully functional web application covering all modules listed in Section 4
2. Guest App and Staff App published/ready for iOS and Android
3. Complete source code with version control history (e.g., Git repository access)
4. API documentation (Swagger/OpenAPI) for all integrations
5. Database schema documentation
6. Deployment/installation guide for the production environment
7. User manuals for each staff role (Front Desk, Housekeeping, Finance, Admin, etc.)
8. Admin configuration guide
9. Test cases and QA test results/report
10. Training sessions (recorded and/or live) for staff and administrators
11. Warranty/support period as agreed in the contract

9. Testing & Quality Assurance

Every module must be tested before it is considered complete. "Complete" means it passes all the testing types below, not just that the code has been written.

Testing Type	What It Confirms
Unit Testing	Individual functions/components tested in isolation by developers as they build (e.g., does the tax calculation function return the correct amount).
Integration Testing	Testing that modules work correctly together (e.g., a booking made in Reservations correctly appears in Front Desk and Billing).
System/Functional Testing	Testing complete user workflows end-to-end (e.g., full guest journey from booking to checkout and invoice).
User Acceptance Testing (UAT)	Client's actual staff test the system in a safe environment and approve it before go-live.
Performance/Load Testing	Simulating many concurrent users to confirm the system meets the speed and scalability targets in Section 6.
Security Testing	Penetration testing and vulnerability scanning, especially for payment and login areas.
Regression Testing	Re-testing existing features after every new update to make sure nothing that used to work has broken.

10. Deployment, Training & Ongoing Support

10.1 Deployment

- A staging/testing environment (separate from production) must be used for all testing and client review
- Production go-live should happen only after UAT sign-off (Section 9)
- A rollback plan must exist in case go-live reveals a critical issue
- Any existing data (current bookings, guest history) must be migrated and verified before go-live, if applicable

10.2 Training

- Role-based training sessions for each staff group (Front Desk, Housekeeping, Finance, Management, Admin)
- Hands-on practice in the staging environment before go-live
- Recorded training videos and written user manuals for future/new staff

10.3 Post-Launch Support ("Hypercare")

- A defined period (e.g., 2–4 weeks) immediately after go-live with faster response times to catch and fix any issues quickly
- Daily check-ins with the client's operations team during this period

10.4 Ongoing Maintenance & Support (After Hypercare)

Issue Severity	Target Response & Resolution Time
Critical Issue (system down / cannot take bookings)	Response within 1–2 hours, fix or workaround within 24 hours
High Priority (major feature broken, no workaround)	Response within 4 hours, fix within 2–3 business days
Medium Priority (feature partially broken, workaround exists)	Response within 1 business day, fix within 1–2 weeks
Low Priority (cosmetic issue, minor request)	Response within 2–3 business days, scheduled into a future update

Note: These are suggested Service Level Agreement (SLA) targets — final numbers should be confirmed and formalized in the support contract.

11. Assumptions & Dependencies

- The client will provide timely feedback and content (logos, room photos, tax rules, policies) needed during development
- Third-party services (payment gateways, OTAs, SMS providers) will provide API access and any required developer accounts
- The client will designate a point of contact/product owner available for regular check-ins and UAT
- Any change to the modules or features listed in this document after approval will go through a formal Change Request process, which may affect timeline and cost

12. Acceptance Criteria

A module is considered "done" and accepted only when all of the following are true:

1. All features listed for that module in Section 4 are implemented and working as described
2. The module has passed unit, integration, and system testing (Section 9)
3. The client's team has completed UAT for that module without unresolved critical or high-priority issues
4. Relevant documentation (user manual, admin guide) for that module has been delivered
5. The module meets the non-functional requirements in Section 6 (security, performance, etc.)

13. Approval & Sign-Off

By signing below, both parties confirm they have reviewed and agreed to the scope, modules, and terms described in this document.

Party	Date
Client Representative — Name & Signature	Date: _____
Development Team Representative — Name & Signature	Date: _____